Final Project Report: Neural Network Implementation & Image Classification

Steven Oviedo Aristizabal

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1 Best Model Architecture

Across the experiments, the most successful architectures had an equilibrium between capacity and heavy regularization.

- **For the MNIST Dataset:** The best architecture was **Architecture B**. The structure was: `Linear(784 → 256) → BatchNorm1D → ReLU → Dropout(p=0.5) → Linear(256 → 10)`.
- **For the Kaggle Intel Image Dataset:** The best architecture was **Architecture E**. The structure was: `Linear(22500 → 2048) → BatchNorm1D → ReLU → Dropout(p=0.6) → Linear(2048 → 6)`.

2 Training and Validation Curves

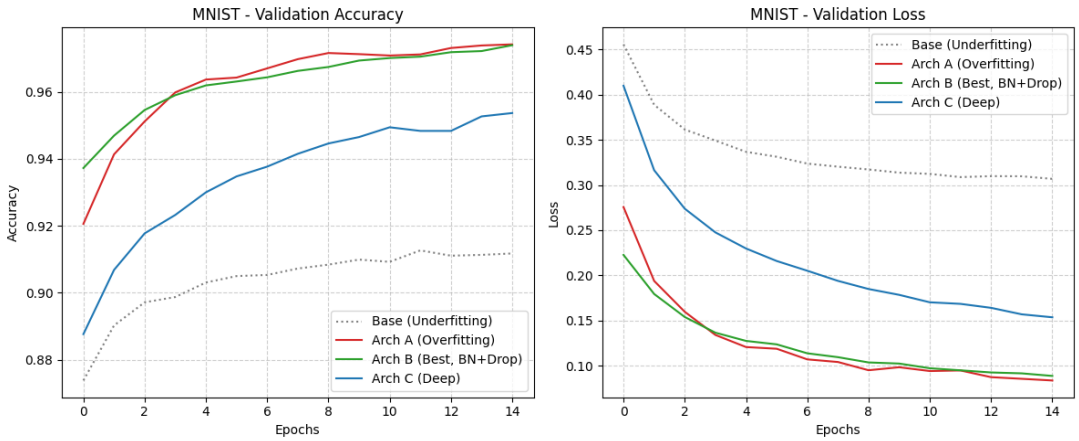

Figure 1: MNIST Dataset: Effect of BatchNorm and Dropout on accuracy and loss.

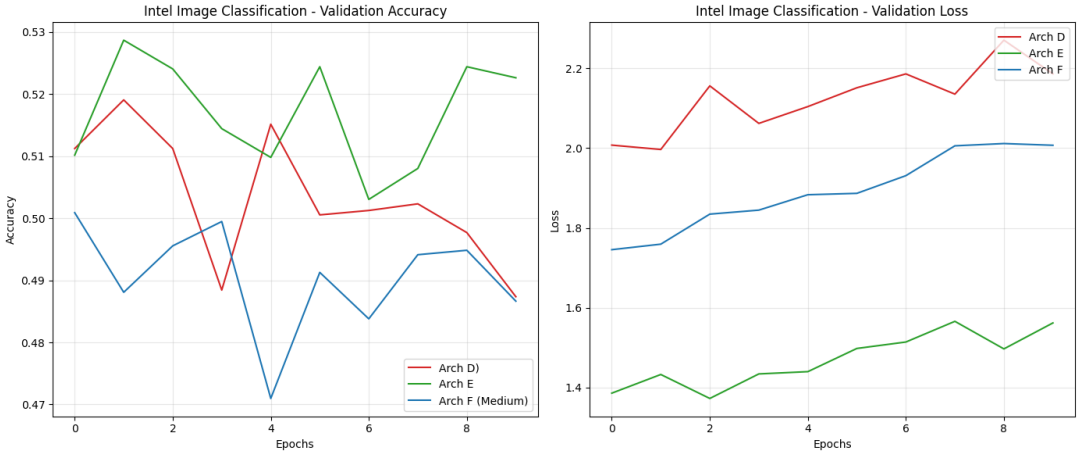

Figure 2: Kaggle Intel Dataset: Comparison of wide vs. deep architectures.

The optimal curves showed the training loss smoothly decreasing while the validation loss stabilized and remained close to the training loss.

Without regularization (as seen in Architectures A or D), the curves displayed a classic mandibular shape—training accuracy approached 99% while validation accuracy flatlined or degraded, indicating severe overfitting. The regularized models successfully kept the gap between Training and Validation accuracy to a minimum.

3 Final Accuracy / Score

- **MNIST (Architecture B):** Reached a peak Validation Accuracy of ∼**97.86%** with an overfitting gap of less than 0.5%.
- **Kaggle (Architecture E):** Reached a peak Validation Accuracy of ∼**52.87%**. While lower overall, this was the highest achieved on the complex 150x150 images using strictly linear layers, keeping the overfitting gap under 28% compared to the massive 40%+ gaps seen in deeper unregularized models.

4 Short Reflection on BatchNorm & Dropout Impact

- **Batch Normalization (BN):** Adding BN proved essential for stability. By normalizing the inputs to each layer, it prevented internal covariate shift and stopped the loss from exploding. This allowed the models to use much higher learning rates (up to 0.05 or 0.1) to converge faster without the gradients collapsing.
- **Dropout:** Dropout was the most critical component for ensuring the models actually generalized to unseen data. Flattening large images (like the 22,500 features in the Kaggle dataset) gives the network millions of parameters, making it incredibly easy for the model to simply memorize the training data. By randomly turning off 50% to 60% of the neurons during training, Dropout forced the networks to learn robust, redundant features rather than relying on specific, brittle pixel patterns.